

Localizing the Origin of Idiopathic Ventricular Arrhythmias From ECG Using a Recurrent Convolutional Neural Network With Attention

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Abstract—Idiopathic ventricular arrhythmias (IVAs) are extra abnormal heartbeats disturbing the regular heart rhythm that can become fatal if left untreated. Cardiac catheter ablation is the standard approach to treating IVAs; however, a crucial prerequisite for the ablation is the localization of the IVA origin. The current IVA localization techniques are invasive, rely on expert interpretation, or are inaccurate. This study aims to develop a deep learning algorithm that can automatically identify the origin of IVAs from electrocardiogram (ECG) signals without the need for expert manual analysis. Our developed deep learning algorithm comprises a spatial feature extraction to extract the most informative features from multichannel ECG data, temporal modeling to capture the evolving pattern of the ECG time series, and an attention mechanism to weigh the most important temporal features and improve the model's interpretability. The algorithm was validated on a 12-lead ECG dataset collected from 334 patients (230 females, age 46 ± 13) who experienced IVAs and successfully underwent a catheter ablation procedure that determined the IVAs' exact origins. It achieved an area under the curve (AUC) of 93%, an accuracy of 94%, a sensitivity of 97%, a precision of 95%, and an $F1$ -score of 96% in locating the origin of IVAs and outperformed existing automatic and semi-automatic algorithms. Our proposed method shows promise toward automatic and noninvasive evaluation of IVA patients before cardiac catheter ablation.

Index Terms—Deep learning, electrocardiogram (ECG), idiopathic ventricular arrhythmia (IVA).

I. INTRODUCTION

PREMATURE ventricular complex (PVC) is defined as a premature heartbeat that occurs when the lower chambers of the heart contract too early [1]. Ventricular tachycardia (VT) is a heart rhythm disorder (arrhythmia) defined as three or more PVCs in a row, at a rate of more than 100 PVC beats per minute. For normal individuals, an occasional period of PVC is not considered a problem and usually does not need treatment [2]. However, PVCs become more of a concern,

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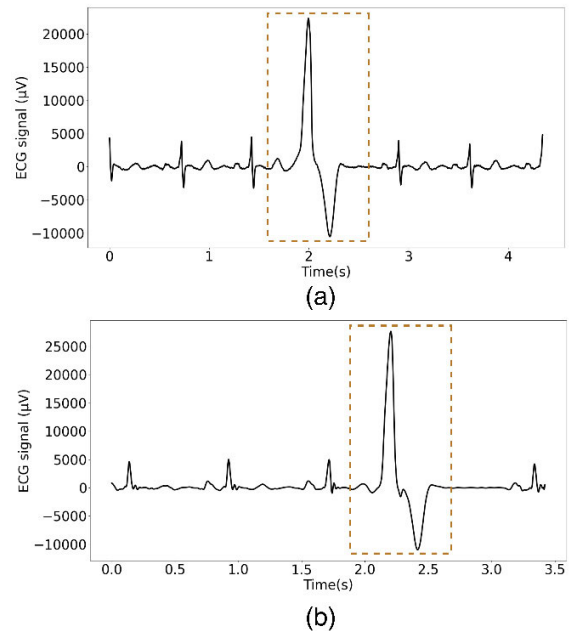


Fig. 1. Illustration of PVC beats. PVC beat originating from (a) RVOT and (b) LVOT.

if they happen frequently or when other heart problems are present. For example, for an individual whose ventricle already squeezes poorly, PVCs may be life threatening. Idiopathic ventricular arrhythmias (IVAs) are those PVCs and VTs that occur for an unknown reason and in the absence of structural heart disease.

Cardiac catheter ablation is usually performed to prevent IVAs and to restore a normal heart rhythm [3]. This procedure consists of two main stages, a diagnostic stage in which the origins of electrical signals causing IVAs can be found, and a treatment stage where the abnormal tissue is ablated (destroyed) through heating or freezing. Thus, it is essential to correctly locate the origin of the abnormal heart electrical activity before ablation. The origin of these abnormal signals is mainly located in either the right ventricular outflow tract (RVOT) or the left ventricular outflow tract (LVOT) [2], [4]. Fig. 1 shows an example of PVC beats in two electrocardiogram (ECG) recordings, one originating from RVOT and the other from LVOT.

Traditional methods for detecting the origins of IVA include electrical mapping [5], substrate mapping [6], and pace mapping [7]. Electrical mapping uses a catheter to record heart

electrograms, but it is limited to patients who can maintain stable IVA throughout the procedure. Substrate mapping identifies ventricular conduction issues by detecting abnormal electrograms, but these signals can be ambiguous and hard to distinguish in scarred tissues. Pace mapping, an alternative to electrical mapping, involves stimulating the heart at various sites and matching the generated electrical pathway to the clinical IVA pattern. These conventional techniques, while effective, are invasive, risky, and costly, demanding significant time and expertise. In contrast, the scientific community is increasingly interested in using ECG as a noninvasive, cost-effective alternative for IVA origin detection [8]. Automated ECG-based methods for IVA source localization reduce the need for invasive procedures and expert intervention, offering a more patient-friendly approach. This shift not only saves time and resources but also minimizes patient discomfort and potential procedural risks, marking a significant advancement in cardiac diagnostics.

Several studies have already proven a strong relationship between the features of ECG and the locations where IVAs stem from [9], [10], [11], and [12], and several approaches have been developed to find the IVAs' origins based on this relationship [13], [14], [15], [16], [17]. For example, in [13], a threshold based on the amplitude of the ECG R wave in lead I was used to identify the IVA origin. In [14], a mathematical model based on S wave and R wave amplitudes was introduced as an index for the identification of the IVA origin. In [15], ECG QRS morphological features were extracted by three electrophysiologists, and an extreme gradient boosting tree classifier was designed for the detection of IVA origins. In [16], a 12-lead ECG dataset was simulated using a computer heart model to train a convolutional neural network (CNN) for the localization of IVA. In [17] a support vector machine (SVM) was designed to locate IVA origins. The use of CNNs was also investigated in [17]; however, the designed network could not outperform the conventional SVM model.

Given that automatic detection of morphological features in an abnormal ECG is challenging, most IVA localization methods rely on manual ECG morphological feature extraction and analysis [10], [13], [14], [18], [19]. Among automatic IVA localization algorithms, some rely on simulated data that may not fully represent the real ECG [16] while others rely on conventional machine learning algorithms that are not capable of fully extracting the ECG spatial and temporal information [17]. There is therefore a significant demand for a fully automatic approach that can detect the origin of IVA from the complex spatial-temporal pattern of ECG multichannel data without the need for any manual analysis. Among different automatic techniques for characterizing ECG multichannel data, deep learning has shown greater potential and more accurate results in various applications [20], [21].

In this article, we propose an end-to-end deep learning-based framework trained on a real-world dataset to automatically localize IVA origins. Our feature extraction and training phases are end-to-end which gives the algorithm the ability to automatically extract the most informative features correlated with IVA origins. Our deep learning framework relies on a combination of CNNs and recurrent

TABLE I
PATIENTS CHARACTERISTICS

Characteristics	All	RVOT	LVOT
Patients, n	334	257	77
Age, Mean $\pm$ SD, year	46.1 $\pm$ 13.1	47.5 $\pm$ 13.4	46.2 $\pm$ 16.5
Male, n	104	65	39
Frequent PVC, n	325	251	74
Sustained VT, n	9	6	3

neural networks (RNNs) to capture the spatial information contained within the ECG multichannel data and the temporal information contained in the ECG time pattern. An attention mechanism is employed to further focus the algorithm on the most important segments of the ECG signal and improve the model's interpretability. The proposed framework is validated on a dataset of 334 and is benchmarked against state-of-the-art deep learning algorithms.

II. METHOD

A. Dataset

This study was conducted on a publicly available 12-lead ECG dataset collected from 334 patients who experienced IVA and successfully underwent a catheter ablation procedure to validate the exact origin of IVAs [2].

The dataset has been collected under the auspices of Chapman University and Ningbo First Hospital of Zhejiang University. The institutional review board of Ningbo First Hospital of Zhejiang University has approved this study and has allowed the data to be shared publicly after de-identification.

The average length of ECG recordings in the dataset is 10.53 ± 3.58 s and the sampling rate is 2000 Hz. Of the 334 patients, 77% are classified as RVOT and 23% as LVOT. Participants' characteristics are provided in Table I.

B. Problem Formulation

Determining the source of IVA can be framed as a time-series classification challenge. This involves developing a robust model capable of extracting meaningful insights from ECG recordings of varying lengths, and accurately categorizing each record into the appropriate class. Our strategy entails transforming this challenge into a deep-learning framework. In this process, the input consists of ECG signals with varying lengths. Initially, these signals are zero-padded and subsequently fed into the model. The model then produces a binary output, indicating the classification as LVOT or RVOT.

The main goal of this deep learning framework is to minimize the binary cross-entropy loss, as it effectively considers the predicted probabilities of each class, enabling a more nuanced evaluation of the model's performance [22], [23]. This minimization aimed to reduce the discrepancy between the

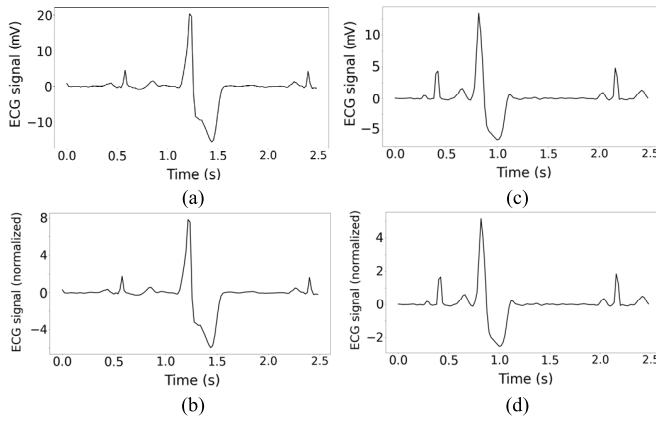


Fig. 2. ECG normalization examples. (a) LVOT-origin PVC beat pre-normalization; (b) post-normalization; (c) RVOT-origin PVC beat pre-normalization; and (d) post-normalization.

reference labels and the model's predictions. Mathematically, this is represented as follows:

$$\text{Loss} = -\frac{1}{N} \sum_{i=1}^N (y_i \cdot \log y'_i + (1 - y_i) \cdot \log(1 - y'_i)) \quad (1)$$

where y_i is the reference label, y'_i is the model output for the i th ECG record, and N is the total number of available ECG records. The binary cross-entropy loss was chosen to minimize the distance between two predicted and actual probability distributions.

C. Preprocessing

As our dataset contained ECG signals with different lengths, we padded each ECG signal up to 25 s by adding zeros to the end of each signal as the longest signal has 25 s length. Padding ECG signals allows the application of deep learning algorithms that require input samples with equal lengths. ECG signals were then lowpass filtered at 25 Hz with a fourth-order Butterworth filter to remove high-frequency noise and downsampled to 50 Hz to reduce the input data dimension. ECG segments were then normalized by removing the mean and scaling to unit variance. The goal of normalization was to transform features to be on a similar scale. This improves the performance and training stability of the model [24]. Fig. 2 illustrates an example of a signal both before and after undergoing normalization.

D. Model Architecture

The extensive capability of deep neural networks (DNNs) in predictive modeling of complex physiological processes, including diverse cardiovascular signals such as ECG [25], [26] and photoplethysmography (PPG) [27], [28], [29], has been firmly proven. Given the promising performance of these networks, we have implemented a novel framework for a DNN, designed exclusively for the analysis of multichannel ECG data. Our architecture effectively combines CNN, long short-term memory (LSTM), and an attention mechanism in a synergistic manner. The architectural framework comprises four discrete stages of processing and learning: spatial feature

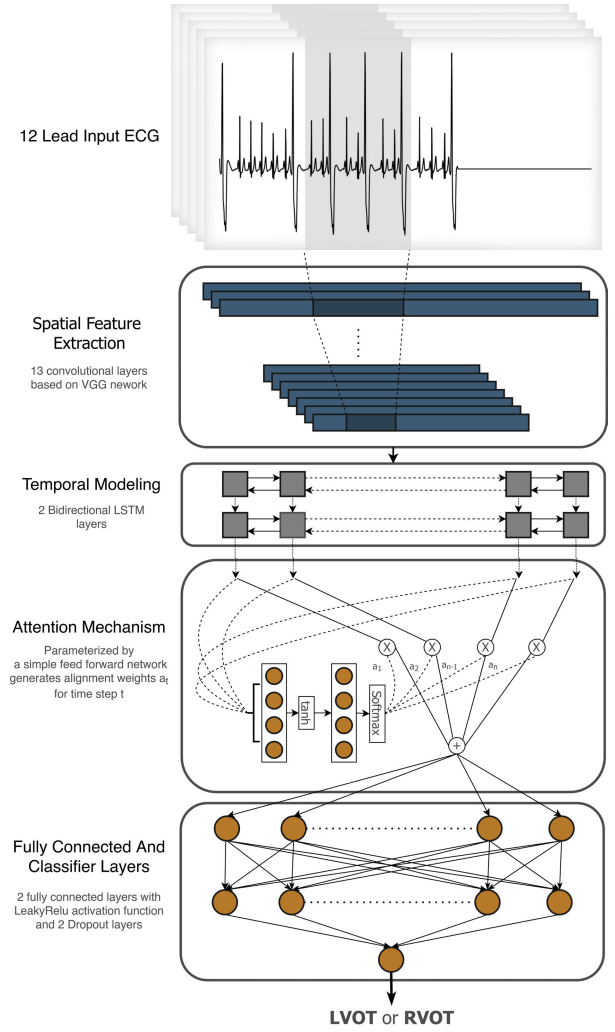


Fig. 3. Block diagram of our proposed model for localizing the IVA origins.

extraction which is a technique that facilitates the integration of spatial information from many channels of ECG data. This integration enhances the network's ability to accurately collect and represent the inter-channel interactions within the data. The incorporation of LSTM in temporal modeling enables a thorough comprehension of temporal dynamics present in ECG signals, which is crucial for achieving precise predictive modeling [30]. The attention mechanism equips the neural network with the capability to dynamically weigh and prioritize specific time steps within the ECG signal, thus focusing computational resources on the most relevant features that inform the diagnostic task at hand. Finally, fully connected layers are employed, wherein the spatial and temporal features that have been extracted are merged and employed to automatically establish the association between the ECG signals and their correlation with the origin of IVA, as depicted in Fig. 3. The proposed architecture effectively captures spatial and temporal information from multichannel ECG signals, offering a reliable framework for automatically modeling their complex association with the origin of IVA.

1) *Spatial Feature Extraction*: In this study, we utilized a CNN, to extract the most relevant features from ECG. CNNs are renowned for their proficiency in hierarchical spatial

feature extraction in the context of 2-D images. The initial layers of a CNN are adept at capturing low-level features, such as edges, while subsequent layers amalgamate these low-level features to construct more abstract representations of the data [31], [32]. It is worth emphasizing that although CNNs are often associated with their effectiveness in the realm of 2-D images, their applicability extends to biological signals, as evidenced by numerous studies highlighting their proficiency in extracting features from signals like ECG [33], [34]. In this endeavor, we designed a 13-layer CNN architecture based on the Visual Geometry Group (VGG) structure [35]. The VGG architecture has been validated as a potent spatial feature extractor for 1-D signals [25]. Our custom-designed CNN encodes the 12 ECG channels, each with a size of 1250, into 512 channels of size 78. The resultant encoded data retained critical spatial information, which was subsequently subjected to further processing through the temporal modeling module and the attention mechanism for the purpose of IVA localization.

2) *Temporal Modeling*: To effectively capture temporal dependencies within the input ECG sequence, a dedicated temporal modeling module has been incorporated. While CNN layers excel at spatial feature extraction, they are limited in their ability to fully grasp temporal characteristics [25]. To address this limitation, the temporal modeling module comprises two stacked bidirectional LSTM (BiLSTM) layers [36], [37]. The utilization of an RNN, specifically the BiLSTM layers, holds paramount significance due to their exceptional ability to apprehend and encapsulate temporal dependencies within the ECG time sequence. In the context of our ECG analysis, where temporal information is assumed to have a pivotal role, the RNN becomes indispensable for comprehending the intricate and dynamic changes in the heart's activity over time. Each BiLSTM layer is composed of two LSTM layers: one that processes the input time series in a forward time progression and another that processes them in reverse. With 64 LSTM units per BiLSTM layer, these layers receive the 78 encoded samples from the spatial feature extraction output sequentially over 78-time steps. During each time step, the temporal model analyzes all 512 channels of data. After processing each time step, the module generates two vectors, each of size 64, referred to as BiLSTM states. These vectors encapsulate the extracted temporal information from the sequence up to the current time step, with one corresponding to the forward LSTM layer and the other to the backward LSTM layer.

3) *Attention Mechanism*: In this study, we employed an attention mechanism inspired by a similar approach introduced in [38]. The attention mechanism is a powerful tool that can be utilized to automatically assign weights to each ECG time step based on its importance in localizing the origin of the IVA. This is achieved through the use of two fully connected layers with learnable parameters, which are responsible for scoring each time step according to its significance. During the training process, the weights of these fully connected layers are updated, allowing the attention mechanism to learn from the training ECG data to identify patterns that enable it to distinguish the critical parts of the signal. Utilizing an

attention mechanism following the BiLSTM layers confers several benefits to the model's performance and interpretability. By calculating a weighted average of BiLSTM states across all time steps, the attention mechanism enables the model to capture and prioritize information from various temporal contexts. This dynamic focus on different segments of the input sequence is especially valuable when dealing with sequential data like ECG signals. The attention mechanism's ability to assign different importance scores to each time step promotes adaptability. It ensures that the model pays more attention to relevant patterns and disregards noise or less informative portions of the data. This selective attention not only enhances the model's accuracy but also improves its ability to interpret and provide insights into the underlying patterns in the ECG signals.

4) *Fully Connected and Output Layers*: The output from the attention mechanism was further processed through a sequence of neural network layers to create a robust feature representation and facilitate the final decision-making process.

Initially, the attention output was directed into two fully connected layers, each comprising 256 nodes and LeakyReLU activation functions with a 0.01 negative slope. These fully connected layers played a pivotal role in consolidating the information gleaned from the preceding layers, effectively amalgamating the diverse features extracted throughout the network. By doing so, they crafted a more abstract and comprehensive feature representation, a critical step in making the ultimate classification decision. To prevent overfitting, which could lead to suboptimal model generalization, two dropout layers were introduced into the architecture. These dropout layers were configured with a dropout rate of 0.2, strategically withholding a fraction of the neural network's nodes during training to encourage robustness and reduce the risk of excessive reliance on specific features. The final output layer features a sigmoid activation function, which introduces nonlinearity into the model, enabling it to learn complex decision boundaries for effective differentiation between LVOT and RVOT [39]. This function also provides a probabilistic output between 0 and 1, aligning with the binary classification where 1 indicates RVOT and 0 signifies LVOT, thus capturing the intricate relationships necessary for this task.

E. Performance Assessment

To rigorously assess the generalizability of the proposed methodology on previously unseen data, a tenfold cross-validation strategy was employed. This approach allows for reliable performance evaluation across different folds. In each fold, the dataset was partitioned such that 80% of the data were dedicated to training the model. In addition, 10% of the data served as a validation set, utilized for tuning the model's hyperparameters to achieve optimal performance. The remaining 10% of the data were reserved to gauge the performance of the meticulously tuned model on completely new and unseen data instances. This tenfold cross-validation procedure was meticulously repeated, encompassing every individual's data exactly once in the test set, thereby ensuring comprehensive evaluation [40]. To facilitate model optimization, the Adam optimization algorithm was employed to minimize the binary

TABLE II
PROPOSED MODEL HYPERPARAMETERS, THEIR SEARCH INTERVALS, AND THEIR OPTIMUM VALUE

Hyperparameter	Search interval	Optimum value
Number of convolution layers	3 to 15	13
Number of BiLSTM layers	1 to 3	2
Number of LSTM units	32 to 512	64
Number of nodes in the attention layer	16 to 128	64
Number of fully connected layers	1 to 3	2
Number of nodes in the fully connected layer	128 to 1024	256
Learning rate	1×10^{-5} to 1×10^{-2}	5×10^{-5}
Batch size	32 or 64	32
Sampling frequency	30 to 300	50
Low pass filter cutoff frequency	15 to 150	25

cross-entropy loss function, which is essential for training the model accurately.

To pinpoint the best-suited hyperparameters for our model, a random search was conducted across various hyperparameter combinations. This systematic exploration allowed us to identify the optimal configuration that yielded the highest validation performance. The comprehensive list of model hyperparameters can be found in Table II, offering transparency into the intricacies of our model's configuration.

III. RESULTS

We evaluated our proposed algorithm for localization of the origin of IVAs on data collected from 334 individuals (see Table I) using a tenfold cross validation.

The optimum model hyperparameters selected based on the validation results are listed in Table II. Our experiments revealed a DNN structure with 13 convolution layers, 2 BiLSTM layers with 64 LSTM units, an attention layer with 64 nodes, and 2 fully connected layers with 256 nodes and LeakyReLU activation functions with negative slopes of 0.01 leads to the best generalization performance. The optimum training parameters included a batch size of 32 and a learning rate of 5×10^{-5} .

The achieved results on the test sets (not seen during training and validation) are reported in Table III. The results are reported for every fold as well as their average value on the whole dataset in terms of the area under the curve (AUC), accuracy (ACC), sensitivity (SE) or recall, specificity (SP), and $F1$ -score (F1). It is observed that depending on the train/validation/test split the localization results vary. For example, the accuracy varied from 84.8% in fold 7 to 100% in fold 3 while the average accuracy over all the folds was 94.3%. A similar pattern is also exhibited for other classification metrics. On average, the proposed method achieved relatively high performance in terms of all the evaluated metrics with AUC of 93.3%, ACC of 94.3%, SE of 97.2%, SP of 84.4%, PPV or 95.4%, NPV of 90.2%, and $F1$ -score of 96.3%.

TABLE III
PERFORMANCE OF THE PROPOSED MODEL IN THE IDENTIFICATION OF IVA ORIGINS USING TENFOLD CROSS VALIDATION. THE RESULTS ARE REPORTED ON THE TEST SET

Fold Number	AUC	ACC	SE	SP	PPV	NPV	F1
1	97.2	96.9	100	85.7	96.2	100	98.1
2	98.2	96.9	96.5	100	100	80.0	98.2
3	100	100	100	100	100	100	100
4	96.5	90.9	95.6	80.0	91.6	88.8	93.6
5	100	100	100	100	100	100	100
6	94.4	93.9	100	66.6	93.1	100	96.4
7	81.5	84.8	84	87.5	95.4	63.6	89.3
8	82.7	93.9	100	66.6	93.1	100	96.4
9	95.1	94.5	100	81.8	92.8	100	96.2
10	84.7	90.9	95.4	81.8	91.3	90.0	93.3
Average	93.3	94.3	97.2	84.4	95.4	90.2	96.3

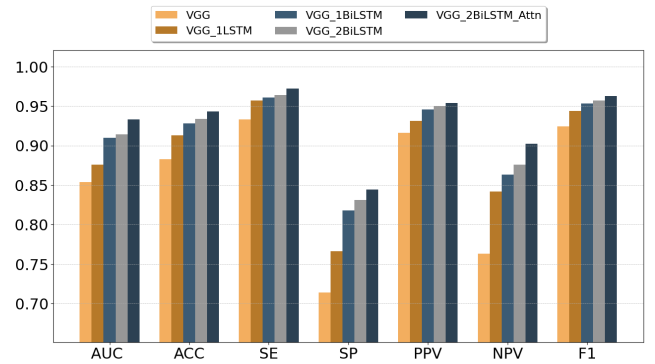


Fig. 4. Impact analysis of various modules in our DNN framework on overall performance.

An ablation study was also performed to evaluate the performance of different modules of our DNN framework on its overall performance. Fig. 4 compares the performance of our DNN framework when using only the VGG model, VGG and one LSTM layer, VGG and one BiLSTM layer, VGG and two BiLSTM layers, and VGG and two BiLSTM layers with the attention mechanism. The AUC, ACC, SE, SP, PPV, NPV, and F1 were 85%, 88%, 93%, 71%, 92%, 76%, and 92% using the VGG network, 88%, 91%, 96%, 77%, 93%, 84%, and 94% by adding a one-layer LSTM, 91%, 93%, 96%, 82%, 95%, 86%, and 95%, by adding a one-layer BiLSTM, 91%, 93%, 96%, 83%, 95%, 88%, and 96% by adding a two-layer BiLSTM, and 93%, 94%, 97%, 84%, 95%, 90%, and 96% by adding an attention mechanism. It is observed that by adding the recurrent architectures the performance is improved and the best results are achieved when combining the VGG network with two BiLSTM layers and the attention mechanism. For example, the AUC was improved by 6% after adding the recurrent layers and by another 2% after adding the attention mechanism.

TABLE IV
COMPARISON OF THE PROPOSED METHOD WITH CUTTING-EDGE DEEP LEARNING ALGORITHMS

Algorithm	Performance						
	AUC	ACC	SE	SP	PPV	NPV	F1
Proposed method	93.3	94.3	97.2	84.4	95.4	90.2	96.3
VGG16	85.4	88.3	93.3	71.4	91.6	76.3	92.4
ResNet18	85.0	82.0	94.1	41.5	84.3	68.0	88.9
EfficientNet	79.0	82.9	94.9	42.8	84.7	71.7	89.5
MobileNet	63.4	76.6	98.0	51.0	77.5	44.4	86.5

Furthermore, we implemented several cutting-edge deep learning models and conducted a comparative analysis to showcase the efficacy of our proposed approach in accurately localizing the IVA origin. The results, which are presented in Table IV, unequivocally demonstrate that our proposed model, which integrates multiple learning stages, outperforms all of the competing models. For instance, VGG16, the closest in performance, recorded an AUC of 85.4%, ACC of 88.3%, and an $F1$ -score of 92.4%. The performance metrics for ResNet18, EfficientNet, and MobileNet further underscore the superiority of our method.

Our proposed method was also compared with three state-of-the-art IVA source localization techniques on the same dataset using the same train, validation, and test sets. The results are reported in Table V. It is observed that the proposed method substantially outperforms the other techniques such as CNN and SVM. For instance, a CNN approach by one study [16] achieved an AUC of 74% and an $F1$ -score of 88%, while another [17] reported an AUC of 68% and an $F1$ -score of 88%. The SVM method showed an AUC of 57% and an $F1$ -score of 82% [17]. The achieved improvements can be attributed to our adopted hybrid deep architecture that extracted both within and between channel information from ECG data, and the attention mechanism that further focused the model's emphasis on important features.

IV. DISCUSSION

Twelve-lead ECG is widely used for diagnosing abnormal cardiac conditions, relying on electrodes placed on the patient's limbs and chest. This noninvasive method is vital for assessing ventricular arrhythmias, yet traditionally requires expert interpretation.

In our study, we leveraged deep learning to identify the origin of IVA from 12-lead ECG data, thereby reducing the reliance on expert analysis. We developed a DNN that integrates a CNN for spatial feature extraction, an RNN for time pattern analysis, and an attention mechanism to highlight significant ECG time steps. This innovative approach effectively correlated the spatiotemporal patterns of ECG data with IVA origins, achieving a notable AUC of 93.3%.

To evaluate our model, we focused on key metrics such as AUC, ACC, SE, SP, PPV, and $F1$ -score. AUC was particularly significant, indicating the model's effectiveness in differentiating between LVOT and RVOT origins. The $F1$ -score, critical in our imbalanced dataset (257 RVOT cases and 77 LVOT

TABLE V
COMPARISONS OF THE PROPOSED APPROACH WITH STATE-OF-THE-ART IVA LOCALIZATION TECHNIQUES. ALL THE METHODS ARE IMPLEMENTED ON THE SAME DATASET USING THE SAME TRAIN, VALIDATION, AND TEST STRATEGY

Study	Model	Performance					
		AUC	ACC	SE	SP	PPV	F1
Yang, T., et al. [16]	CNN	74	80	94	34	83	88
Nakamura, T., et al. [17]	CNN	68	81	97	26	81	88
	SVM	57	71	84	31	80	82
This study	CNN + BiLSTM + Attention mechanism	93	94	97	84	95	96

cases), provided a balanced measure of precision and recall. These metrics collectively affirmed the model's robustness and accuracy in IVA origin detection, underscoring the potential of deep learning in cardiac diagnostics.

Building on the premise of utilizing deep learning for IVA origin detection, it is important to highlight the significance of temporal features in ECG data, such as heart rate and rhythm. These features are pivotal in understanding cardiac conditions, especially since cardiac arrhythmias like IVA are characterized by disruptions in the heart's temporal sequence, which are clinically significant. Incorporating the ability to analyze these temporal aspects into our deep learning model was a critical factor in enhancing its diagnostic accuracy. Our findings reveal that by integrating temporal modeling within our network, there was a notable improvement in all key performance metrics. Specifically, the AUC, ACC, SE, SP, PPV, NPV, and $F1$ -score increased by 6%, 5.1%, 3.1%, 11.7%, 3.4%, 11.3%, and 3.3%, respectively (see Fig. 4). This enhancement underscores the importance of capturing and analyzing temporal information in ECG signals to accurately pinpoint the origin of IVAs, thereby reinforcing the effectiveness of our model in cardiac arrhythmia diagnosis.

We used an attention mechanism to 1) potentially improve localization performance and 2) enhance the interpretability of the algorithm. It was observed that by adding attention to the network, AUC, ACC, SE, SP, PPV, NPV, and F1 are improved by 1.9%, 0.9%, 0.8%, 1.3%, 0.4%, 2.6%, and 0.6%. The assigned weights of the attention mechanism to different parts

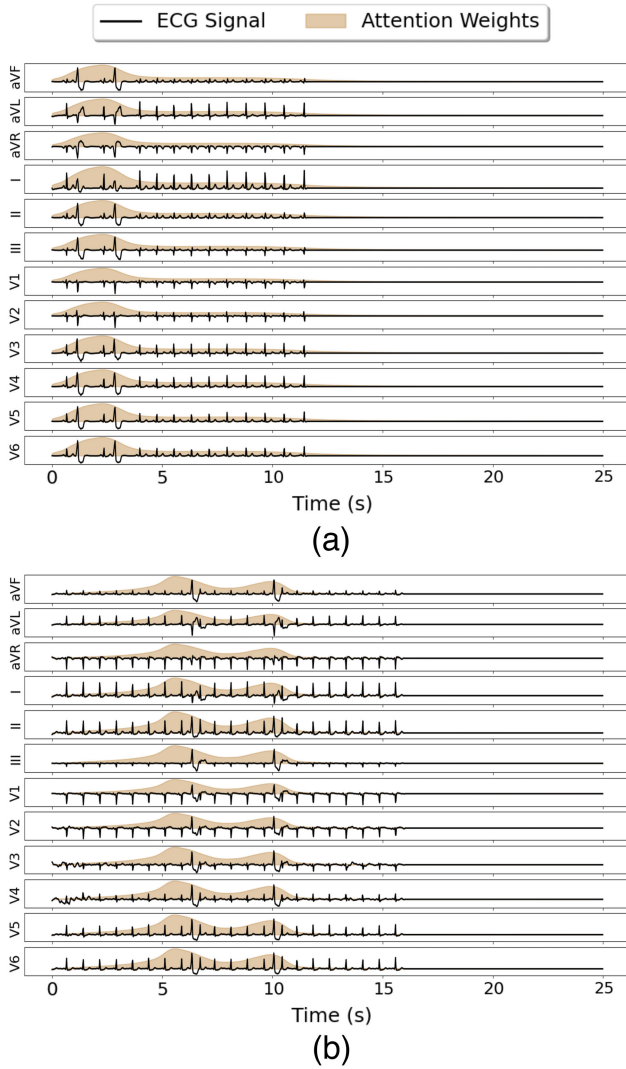


Fig. 5. 12-Lead ECG signal examples for IVA patients. (a) RVOT location. (b) LVOT location. Attention weights are indicated in brown overlaying the ECG signals.

of the ECG signal over time can illustrate their importance in localizing the origin of IVAs. Fig. 5 shows examples of 12-lead ECGs for individuals with RVOT and LVOT. The attention weights are plotted in red on top of the figures. It is observed that attention is focused on specific parts of the input data that are mostly related to abnormal heartbeats. On the other hand, some irrelevant parts of the ECG signal such as the zero-padded segments are completely ignored by the attention mechanism.

The success of our model is not solely due to its strategically designed architecture but also hinges on a rigorous data preprocessing pipeline that includes important steps such as the filtration and normalization of ECG data. Although the American Heart Association (AHA) has established a standard 0.05–150 Hz bandwidth in the recording of 12-lead ECGs [41], [42], our research showed that low-pass filtering the ECGs with a much lower cutoff frequency of 25 Hz can be more beneficial in IVA source localization. We investigated several cutoff frequencies ranging from 15 to 150 Hz and found that a cutoff frequency of 25 Hz returns the best results. The use of

a low cutoff frequency effectively removes the high-frequency noises within the ECG, such as those caused by muscle activities or powerline interference [43]. Furthermore, it allows for downsampling the ECG signals to a lower sampling rate, making computations easier for the model. Besides, based on the findings presented in Table II, our investigation into the optimal sampling frequency of ECG signals suggests that utilizing a 50 Hz sampling frequency yields the most favorable outcomes.

While scaling and normalizing data typically enhance the effectiveness of DNNs [44], it is essential to recognize that these processes may introduce certain changes in the data's characteristics. To comprehensively evaluate the proposed model, we also assessed its performance using unnormalized data. The results obtained from this evaluation demonstrated an ACC of 91.6%, an SE of 95.7%, and an SP of 77.9%. It is worth noting that these metrics are notably lower when compared to the results obtained with normalized data. In addition, in Fig. 2, we provide an illustrative example showcasing PVC pulses both before and after the normalization process. This visualization serves to emphasize that the normalization procedure does not appear to introduce any discernible detrimental changes in terms of morphological or temporal information within the raw data.

Unlike conventional clinical approaches such as electrical mapping [5], substrate mapping [6], and pace mapping [7] which are invasive, our proposed method is solely based on the analysis of noninvasively measured ECG and, therefore, poses no significant risk to the patient. Table VI compares the proposed approach with other conventional and state-of-the-art techniques for localizing the origin of IVA. Unlike manual approaches that rely on the manual extraction and analysis of ECG morphological features such as R and S wave amplitudes and timings [10], [13], [14], [19], our proposed method automatically learns from the spatiotemporal pattern of ECG multichannel data and therefore does not require expert manual analysis. Unlike semi-automatic machine learning techniques that learn from manually extracted ECG features [16], [18], our algorithm automatically extracts and learns ECG features and therefore is less expensive and time-consuming. Among fully automatic techniques, our algorithm is the only algorithm that is validated on real data (unlike [16] which used simulated data), extracts both spatial and temporal features of the ECG signal (unlike [17] that only modeled the ECG spatial information) and is fully validated on all available data using a tenfold cross-validation. The closest results among the existing automatic algorithms are reported in [17] on a dataset of 464 individuals using an SVM model without specific feature extraction. The same approach was implemented and tested on our dataset (see Table VI). Given that the reported results in [17] were calculated on a single hold-out test set, we conjecture that they may not be fully generalizable.

Our method was tested on a modest cohort of 334 individuals with IVA. As the available data were somewhat limited, a cross-validation technique was used to evaluate the generalizability of the model when encountering unseen data. The significance of our cross-validation approach can be observed in Table III, where it is shown that depending

TABLE VI
COMPARISONS OF DIFFERENT ECG-BASED ANALYSIS METHODS FOR LOCALIZING OF THE ORIGIN OF IVA

Category	Author	Method	Limitations
Manual	Cheng et al. (2018) [10]	Defined an index based on R-wave and S-wave amplitudes in right precordial and posterior ECG leads	Relied on experts to identify and quantify R and S waves
	Di et al. (2019)[14]	Defined an index based on R-wave and S-wave amplitudes in leads V ₁ , V ₂ and V ₃	Relied on experts to identify and quantify R and S waves
	Wang et al. (2021) [19]	Defined an index based on R-wave amplitude in lead I	Relied on experts to identify and measure R wave
Semi-automatic	He et al. (2018) [18]	Logistic regression algorithm using R-wave and S-wave features	-Relied on experts to identify and quantify R and S waves -Single hold-out evaluation
	Zheng, Fu et al. (2021) [15]	Extreme gradient boosting tree based on 1,600,800 features extracted from ECG.	-QRS complex was manually extracted before applying machine learning. -Single hold-out evaluation
Automatic	Yang, Yu et al. (2017) [16]	Automatic spatial feature extraction using CNN	-Validated on simulated data (17090) -Only relied on the extraction of spatial features -A separate validation set was not used to adjust model hyperparameters
	Nakamura, Nagata et al (2021) [17]	Automatic spatial feature extraction using CNN	-Only relied on the extraction of spatial features -Single hold-out evaluation -Limited number of patients
		SVM classifier	-Single hold-out evaluation -Limited number of patients (464)
	This study	Attention-based automatic spatial and temporal feature extraction using CNN-BiLSTM	-Limited number of patients (334)

on the selection of the test set, the detection performance can vary. The best-achieved performance (AUC) was up to 100% while by changing the distribution of train and test data the performance was decreased to nearly 80% in some folds. To ensure a fair and unbiased evaluation, we reported the average performance among different folds. This approach is superior to the previous validation methods that held out a specific small portion of data for testing the performance [17]. Using a single hold-out set as the test data can lead to an unreliable (and unstable) performance evaluation.

To provide a more comprehensive evaluation of our proposed method, we conducted additional experiments using two other validation strategies: fivefold cross-validation and leave-one-out cross-validation. In the fivefold cross-validation, the performance metrics were as follows: AUC of 91%, ACC of 91%, SE of 94%, SP of 81%, PPV of 94%, NPV of 79%, and an *F1*-score of 94%. This approach, due to its inherent structure, involved a reduced training size in each fold, which slightly impacted the model's performance. On the other hand, the leave-one-out cross-validation strategy yielded higher results: AUC of 97%, ACC of 94%, SE of 98%, SP of 83%, PPV of 95%, NPV of 91%, and an *F1*-score of 96%. The superior performance of this method can be attributed to its utilization of a larger training set for each validation step. These outcomes highlight an important aspect of our model: its performance is closely tied to the size of the training

data. The reduced training size in fivefold cross-validation led to lower performance, while the more extensive training set in leave-one-out cross-validation resulted in higher accuracy and reliability. As such, we anticipate further improvements in the model's performance with the expansion of the training dataset.

Our study demonstrates the potential of using advanced deep learning techniques, such as our CNN + BiLSTM + Attention mechanism, for noninvasively localizing IVA origins from ECG data, contributing to the evolving landscape of medical AI applications in cardiac diagnostics.

V. LIMITATIONS AND FUTURE WORKS

The present study encountered certain limitations that warrant discussion. First, the study was conducted with a relatively restricted sample size, involving 334 individuals for performance evaluation. In addition, our dataset exhibited an imbalance in terms of both target classes, with 257 patients having RVOT origins out of the 334, and gender distribution, with 230 females out of the 334 individuals. Although we attempted to use data augmentation techniques, such as shifting or modifying the temporal aspects of ECG signals, these efforts did not yield improvements in the localization of IVA.

To address these limitations and enhance the robustness of future investigations, we recommend exploring more advanced data augmentation techniques, including the generation of

synthetic ECG signals using generative adversarial networks (GANs) [45]. Synthetic ECG signals not only expand the dataset but also offer opportunities to balance it in terms of target classes and gender distribution. Furthermore, adopting a semi-supervised learning approach, which leverages both labeled and unlabeled data, can be highly beneficial [46]. Public datasets containing ECGs with PVC or VT, albeit without labeled origins, can be incorporated within an unsupervised framework to potentially boost localization performance [47].

Another noteworthy limitation of our study is the binary classification of IVA origins, which may lack the precision required for certain clinical applications. There is a need for subclassification into more specific sites. However, it is crucial to acknowledge that multiclass classification is considerably more complex and demands a substantially larger dataset to achieve satisfactory performance.

As part of our future work, we propose subjecting our method to testing on a larger, more balanced dataset to enhance its performance and enable more precise localization of IVA origins. By addressing these limitations and exploring advanced techniques, we aim to contribute to the development of more effective and precise clinical tools for IVA diagnosis and treatment planning.

VI. CONCLUSION

We proposed a fully automatic deep learning model to identify IVA origin from noninvasively measured 12-lead ECG. Our algorithm achieved a high localization performance (AUC = 93.3%) and outperformed existing techniques. Our proposed algorithm was based on spatial modeling using a VGG CNN, temporal modeling using a two-layer BiLSTM RNN, and temporal weighting using an attention mechanism. Unlike manual and semi-automatic algorithms, the proposed approach provided end-to-end automatic processing of ECG data without the need for any expert analysis. Given the short-duration ECG signals required for the analysis, our deep learning model provides a cost-effective and low-risk approach toward the identification of the origin of IVAs before cardiac catheter ablation.

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The source code of their developed algorithms is publicly available on GitHub at: <https://github.com/mohammadrezashahsavari/Deep-Learning-Based-Idiopathic-Ventricular-Arrhythmias-Origin-Detection.git>.

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